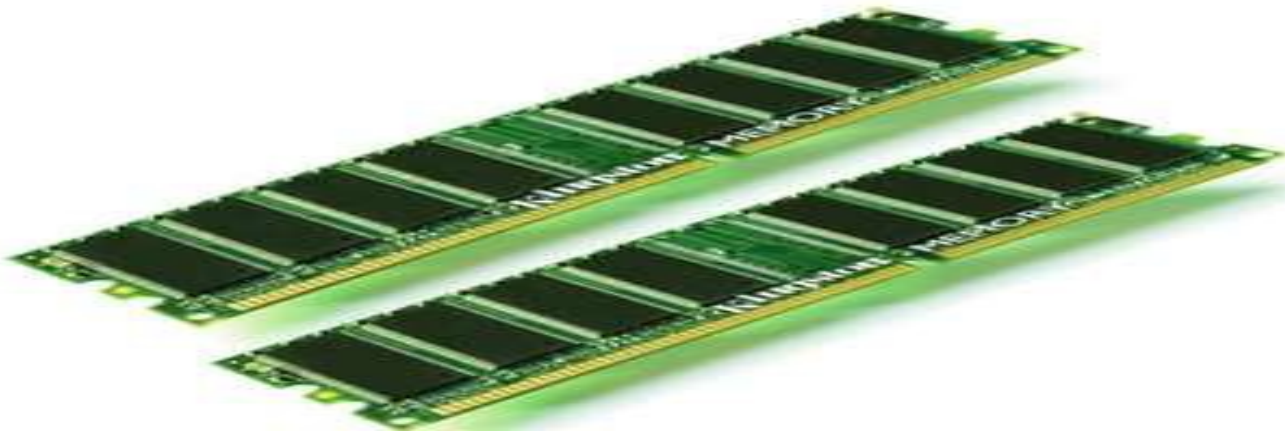


Memory



Memory

- ☑ Memory is the **workspace** for the computer's processor.
- ☑ It is the **temporary storage area** where the programs and data being operated on by the processor must reside.
- ☑ We often call memory RAM.

Memory

- ☑ RAM(Random Access Memory)
 - ☑ is a computer's temporary memory, which exists as chips on the motherboard near the CPU.
 - ☑ It stores data or programs while they are being used and requires power.
 - ☑ provides temporary data storage for the CPU while the computer is operating.
- ☑ The more RAM a computer has, the faster it responds.

Physical Memory Packaging

- ☑ The packaging is simply the entire makeup of a unit of memory,
- ☑ RAM comes in module packaging of **STMM's** and **DTMM's**
- ☑ SIMM's and DIMM's :
 - ☑ are not memory chips.
 - ☑ They are modules.
 - ☑ They are basically just **different ways of packaging** the same silicon memory.
 - ☑ The primary **difference** b/n these two types of modules is in the **number of pins** that they have.

SIMMS /single inline memory module /

- has 2 forms 30 pin(1-16mb) and 72 pin(1-32mb)
- Transmit 4byte(32bit) data at a time

DIMMS/dual inline memory module /

- has 168pins(1-16mb) and 72 pin(1-32mb)
- Transmit 64bit of data at a time
- Capacity from 8mb-256mb

DIMMs and SIMMs

DIMMs

- ☑ have twice as many pins compared to comparable SIMMs.
- ☑ Two DIMM modules can also be paired to upgrade a 128-bit memory system.
- ☑ DIMMs built with high-speed memory chips also have more pins for power and ground connections.

SIMMs

- ☑ have a 32-bit memory bus width, while DIMMs offer double that at 64 bits.
- ☑ Two SIMM modules can be paired to upgrade 64-bit memory systems.
- ☑ Although SIMMs, unlike DIMMs, are outdated, they are still used to upgrade memory on older 32-bit laptops, PCs, servers, printers and workstations.

Types of RAM

- ☑ The two main forms of RAM are **Static RAM (SRAM)**, **Dynamic RAM (DRAM)**.
- ☑ **SRAM**
 - ☑ No refresh
 - ☑ **faster** and more reliable
 - ☑ more expensive
 - ☑ often used as cache memory for the CPU
 - ☑ Do not have to be re-energized as often as DRAM
 - ☑ requires less power than DRAM

Types of RAM

☑ DRAM

- ☑ Refresh
- ☑ is slower
- ☑ Must be re-energized constantly.
- ☑ is **less expensive** to produce than static RAM,
- ☑ is the **predominant** form of computer memory used in modern computers.

Cont..

Both static and dynamic RAM are considered *volatile*, as their state is lost or reset when power is removed from the system.



Read-only Memory (ROM)

- ☑ ROM is a type of "built-in" memory that is capable of holding data and having that data read from the chip, but cannot be written to.
- ☑ ROM chips come from the manufacturer with programs already burned in, or stored.
- ☑ Unlike RAM, ROM is non-volatile which means it keeps its contents regardless if it has power or not.

Types of ROM: Programmable ROM (PROM)

- ☑ This is basically a blank ROM chip that can be **written** to, but **only once**.
- ☑ That is , it can only be changed/updated once.
- ☑ It is much **like a CD-R** drive that burns the data into the CD.
- ☑ Some companies uses special machinery/device to write PROM's **for special purposes**.

Types of ROM: Erasable Programmable ROM (EPROM)

- ☑ This is just like PROM, except that you can erase the EPROM by shining a special **ultra-violet light** into a sensor.
- ☑ Doing this wipes the data out, allowing it to be rewritten.

Types of ROM: Electrically Erasable Programmable ROM (EEPROM)

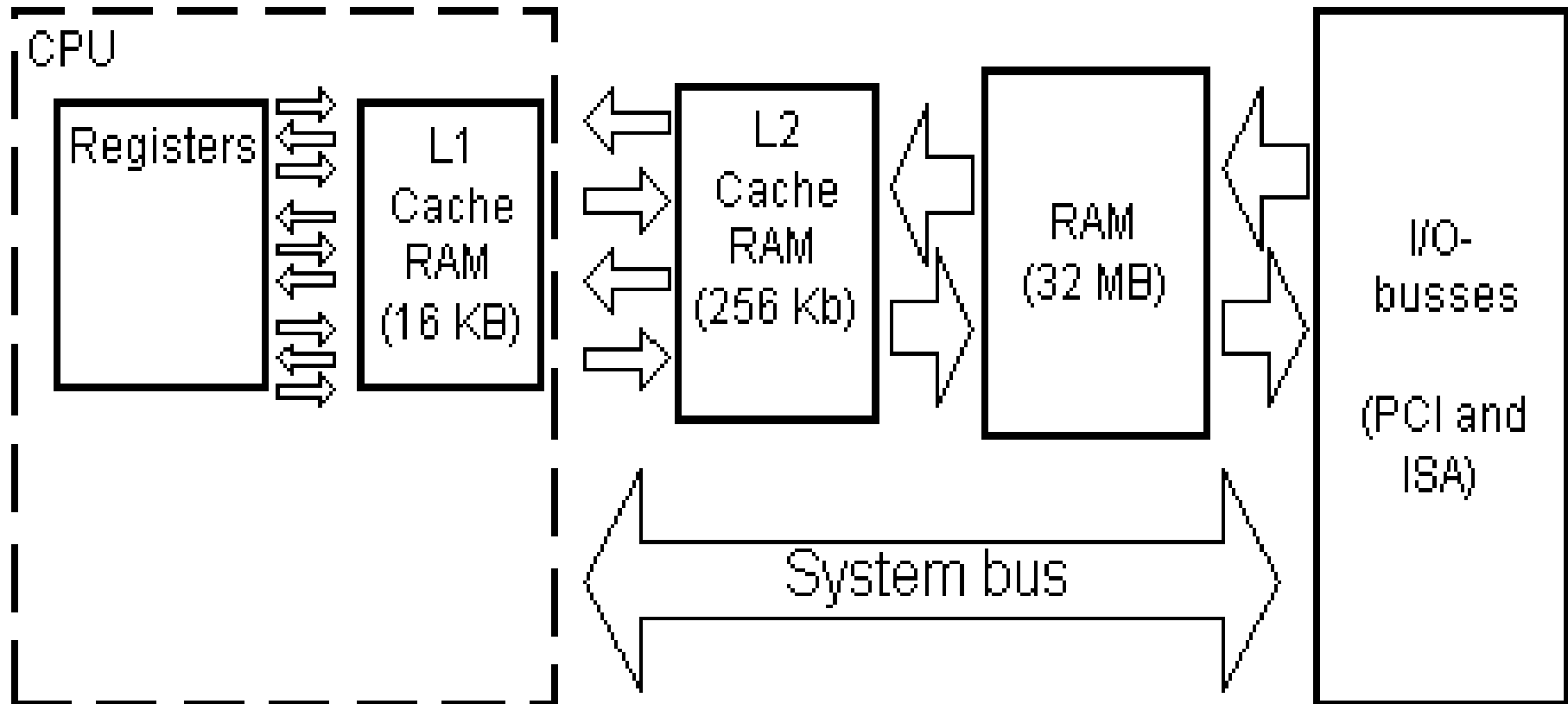
- ☑ Also called **flash BIOS**.
- ☑ This ROM can be rewritten through the use of a special **software program**.
- ☑ Flash BIOS operates this way, allowing users to upgrade their BIOS.
- ☑ **BIOS** is a chipset to which the BIOS manufacturer has written a program code with software known as **firmware**.

Cache Memory

- ☑ The CPU must deliver its data at a very high speed. The regular RAM can not keep up with that speed.
- ☑ Therefore, a special RAM type called cache is used as a buffer - temporary storage.
- ☑ To get top performance from the CPU, the number of outgoing transactions must be minimized.
- ☑ Cache RAM becomes especially important in clock doubled CPU's, where internal clock frequency is much higher than external.

- ☑ Then the cache RAM enhances the "horsepower" of the CPU, by allowing faster receipt or delivery of data.
- ☑ Beginning with 486 processors, *two layers of cache are* employed.
- ☑ The fastest cache RAM is inside the CPU. It is called **L1 cache**.
- ☑ The next layer is the **L2 cache**, which are small SRAM chips on the system board.

Cache Memory



Upgrade Memory

- ☑ If your desktop PC has slowed down, then memory (or the need for more or not enough of it)) may be the culprit. Upgrading memory is the easiest way to modify your computer's speed.
- ☑ The individual memory sticks hold from 128 MB RAM up to 8 GB.
- ☑ The first thing you need to figure out is how much memory and how many empty memory slots you actually have in your system.

Steps

1. Get a new memory stick, then turn off the computer and the power supply and unplug your computer.

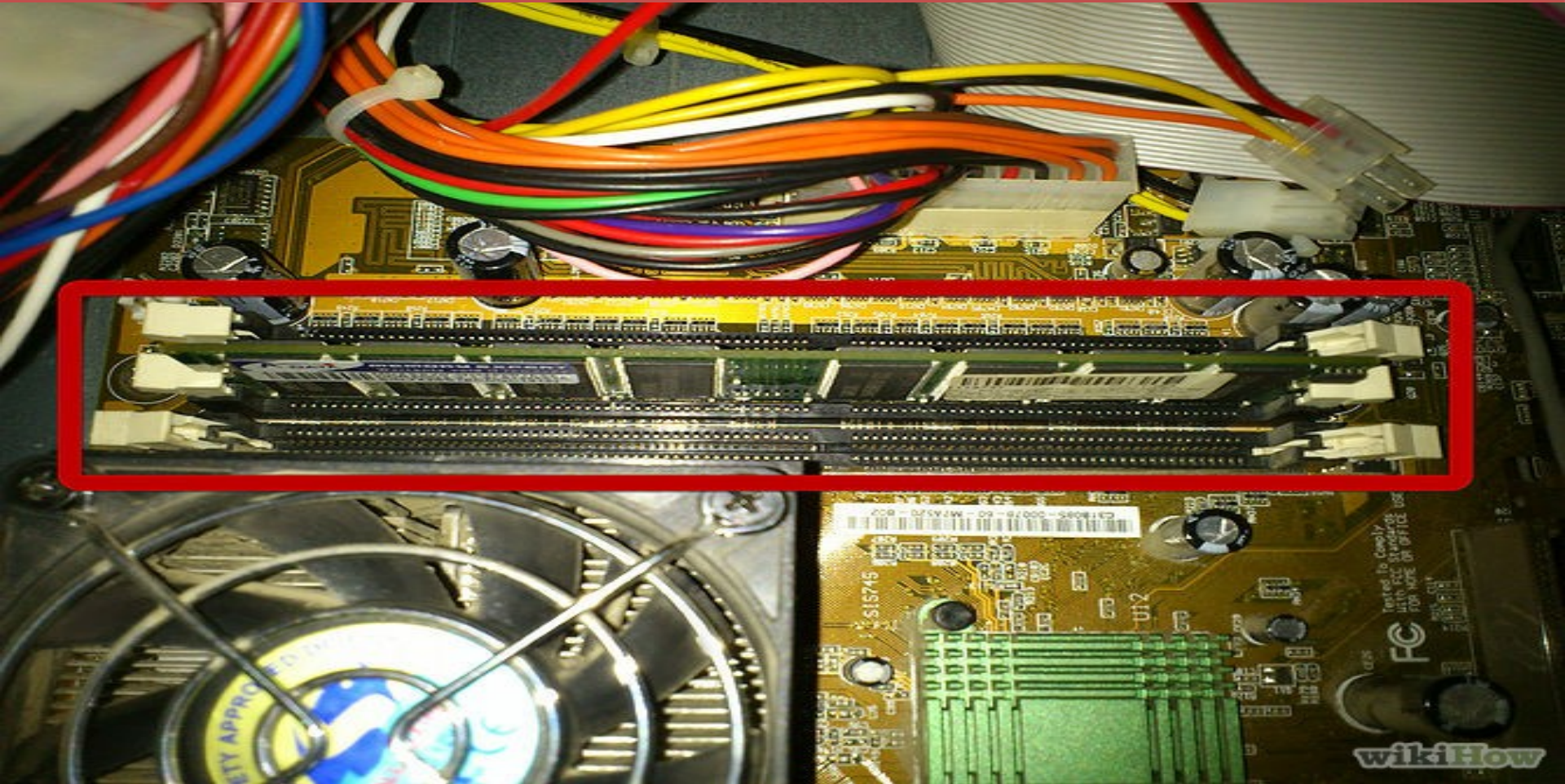


2. Open up your computer by removing the screws in the back. (though some computers have a latch which is pushed)

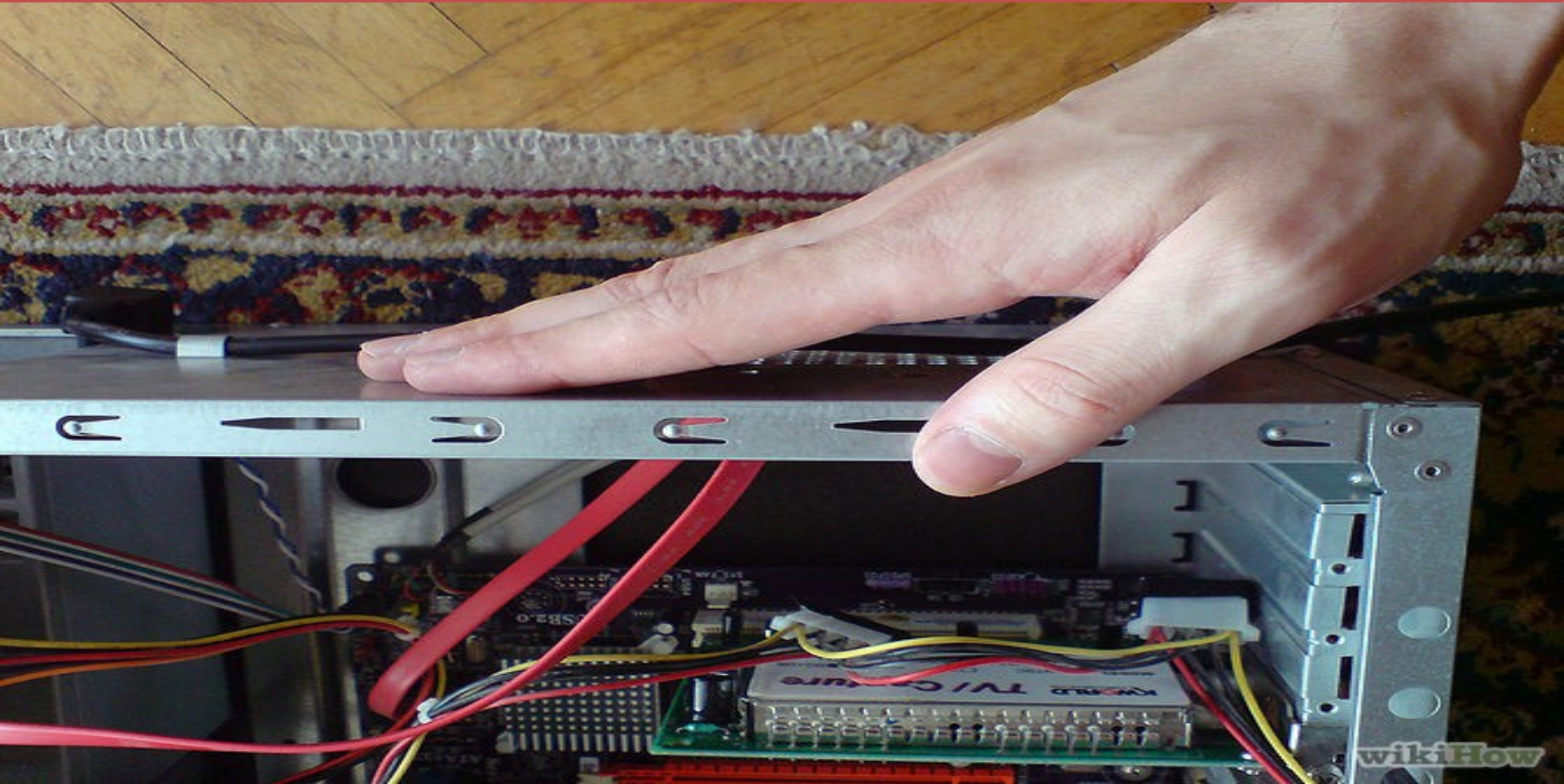


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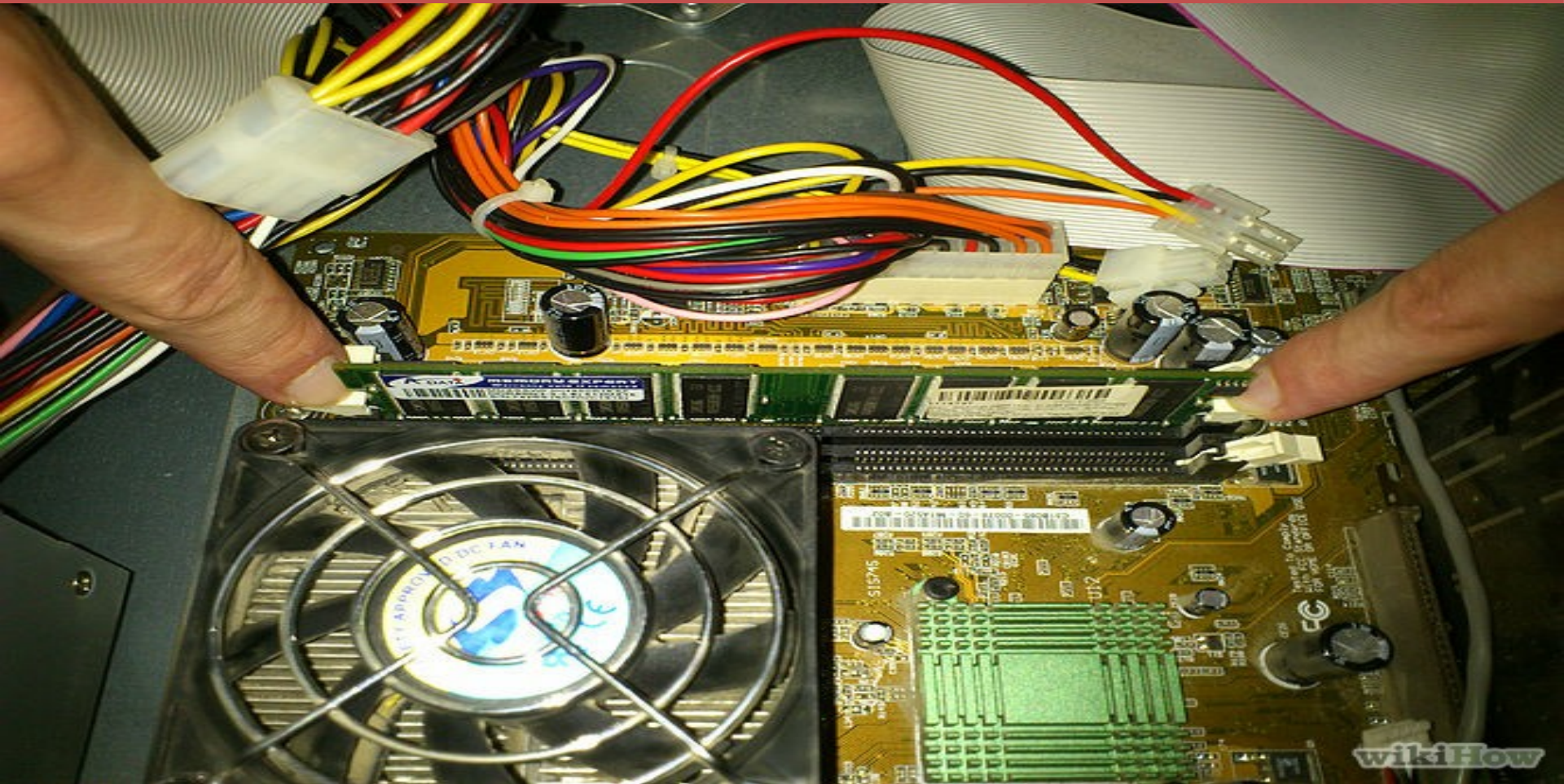
3. Locate the memory cards that are already in your computer. They are usually long and thin, and rectangular in shape.



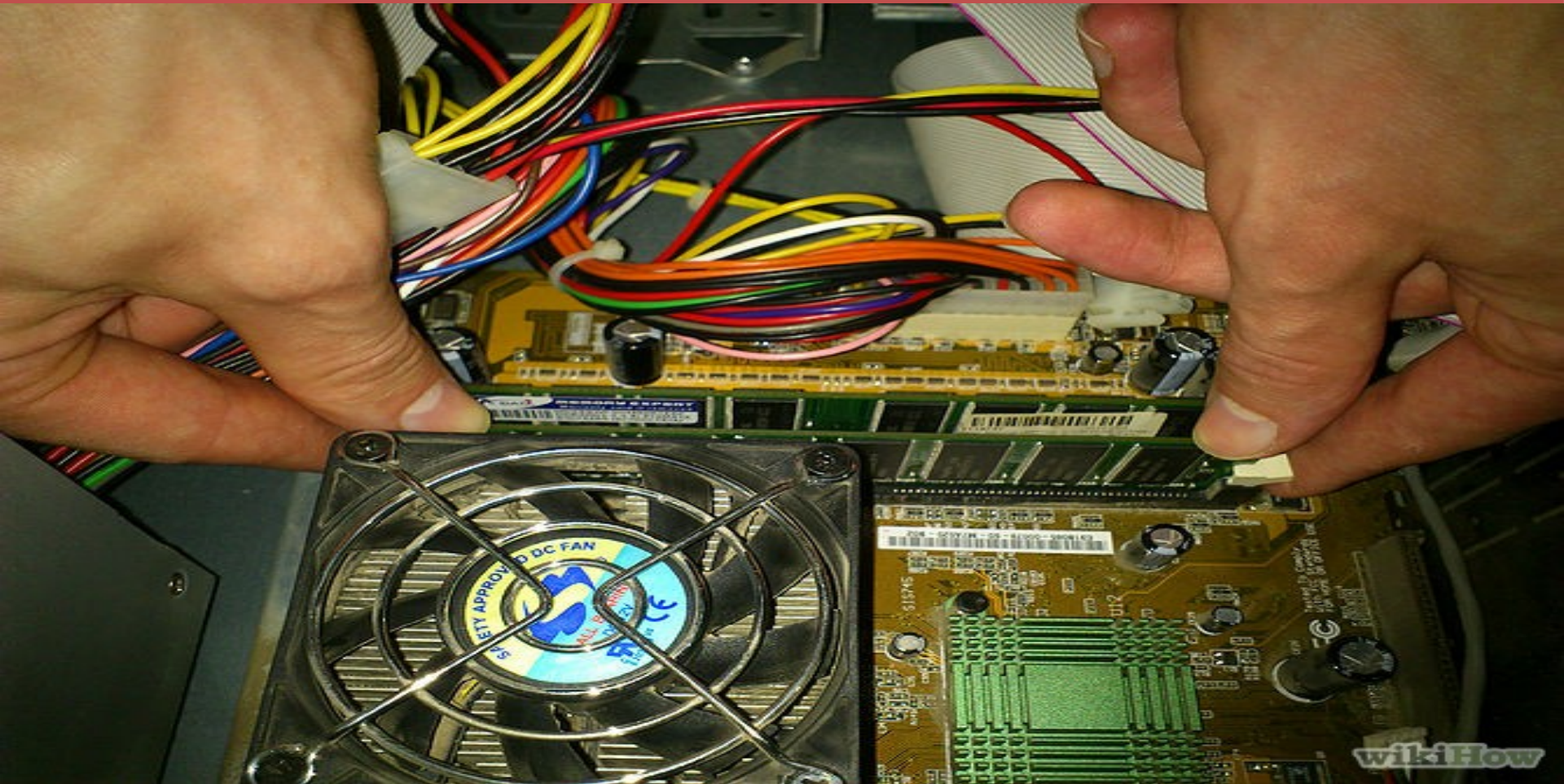
4. Touch the metal computer case with your hand in more than one place to discharge static electricity from your body. This static can damage the memory chips.



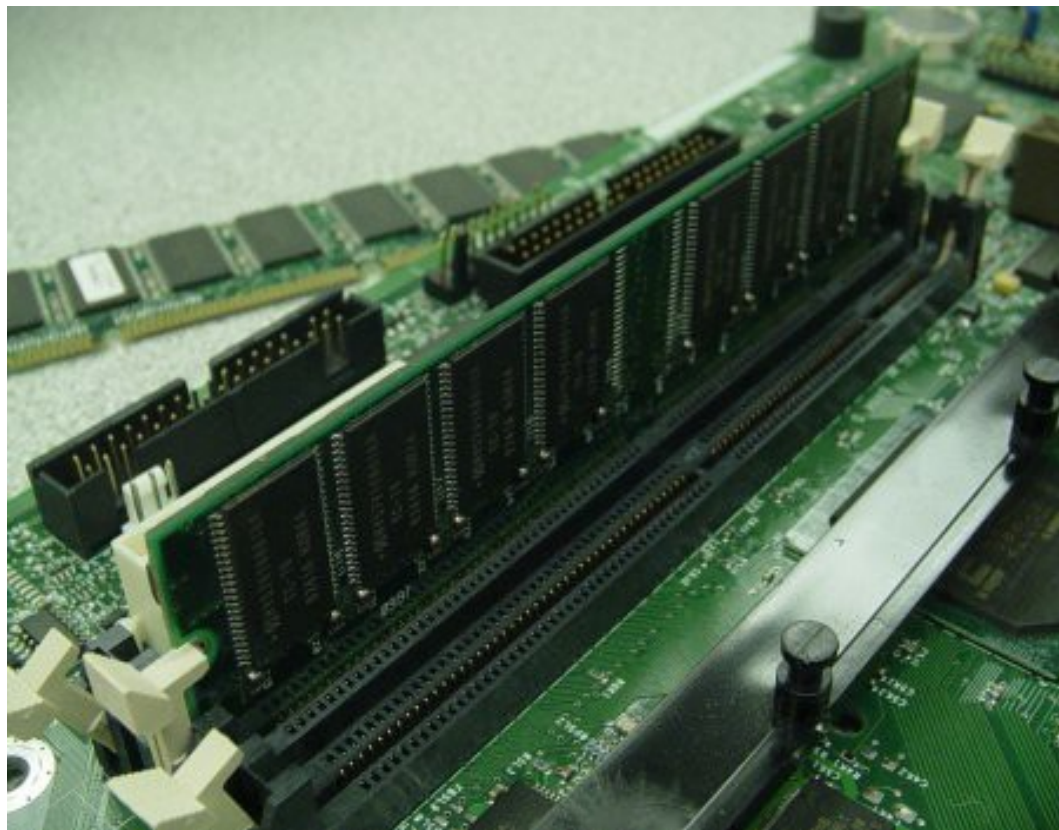
5. Push the two white tabs down to release the current memory and pull out the chip.



6. Line up the a new chip and push it into place. Make sure it is in the whole way. A clicking sound can usually be heard when the memory is secure.



RAM on the slot



Memory Problems and Trouble shootings : Problems

- ❑ No Display and blank screen with a long multiple short beep
- ❑ Computer always hang during the boot process.
- ❑ The installation setup interrupts before confirmation.
- ❑ System refuse to work another task until you close window's
- ❑ Slow down in operation or reboot
- ❑ System freezes and memory size error, memory parity error displayed.
- ❑ Any error starting with 2 usually indicates memory related problems
- ❑ Create the ghost and crashes images

Solution for Memory

- ☑ Remove dust from the memory bank and the memory contact edge.
- ☑ Check the compatibility of the memory interface.
- ☑ Remove dust from the slot.
- ☑ Check the memory installed correctly(or RAM absent)
- ☑ Check the functionality of RAM and make sure the memory slot not defective.
- ☑ Check the capacity of RAM
- ☑ Else use the replacement

Thank You

End of The Chapter